PROJECT REPORT

Student Record and Academic Management System

Name: Shreyasi Ajay Deshmukh

PRN:2126UDSF1049

Division: A

Year: FY

Department: Artificial Intelligence & Data Science

University: Sanjivani University

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Problem Statement

Managing student records by hand through registers, notebooks, spreadsheets, or separate files can be slow and challenging. Finding student details, updating information, recording marks, and working out total marks, averages, percentages, and grades all take time and may result in mistakes.

As the student population grows, keeping records organized becomes more difficult. Information can be lost, duplicated, or updated incorrectly, making it harder to maintain accurate records and track student progress.

The proposed Student Record and Academic Management System offers a straightforward way to organize student details and academic information. Users can add, search, update, and view student records, enter marks for individual subjects, and automatically calculate averages and grades.

This project aims to examine the challenges of managing records manually, determine the system's requirements, and design an effective software-based solution ahead of implementation.

Key Requirements

In short: The system will provide basic student record management, searching, information display, and automatic academic performance calculations.

Feature	Purpose
Add Student	Add a new student's personal information and subject marks.
	Find a student using their Roll Number or PRN.
View Student Details	Display complete student information, including academic marks.
Calculate Average Marks	Calculate the student's average marks from all subject marks.
Calculate Grade	Determine the student's grade based on their average marks.
Exit	Safely exit the system.

Inputs and Outputs

Note: The Student Record and Academic Management System takes student information such as Roll Number, Name, PRN, Class, Division, subjects, and marks as input. The system stores and searches student records using Roll Number or PRN. It calculates the total marks and average marks and determines the student's grade based on predefined grading rules. The system displays student details, successful record confirmation, search results, or "Student Not Found" when no matching record exists.

Input	Process	Output
Student details / information	Store student	Confirmation
Roll Number / PRN	Search the student	Student details / record
Subject marks	Calculate total marks	Total Marks
Subject marks	Calculate average marks	Average Marks
Average marks	Apply grading rules	Grade

Algorithm

Algorithm 1 – Add Student Record

1.Start.

2.Enter Roll Number, Name, PRN, Class, and Division.

3.Enter marks for all required subjects.

4.Store the student details and marks.

5.Display "Student Added Successfully".

6.Stop.

Algorithm 2 – Search Student Record

1.Start.

2.Enter the student's Roll Number.

3.Search the student records.

4.Check whether the Roll Number exists.

4.If found, display the student's details.

5.Otherwise, display "Student Not Found".

6.Stop.

Algorithm 3 – Calculate Academic Result

1.Start.

2.Enter marks for all required subjects.

3.Calculate Total Marks.

4.Calculate Average Marks.

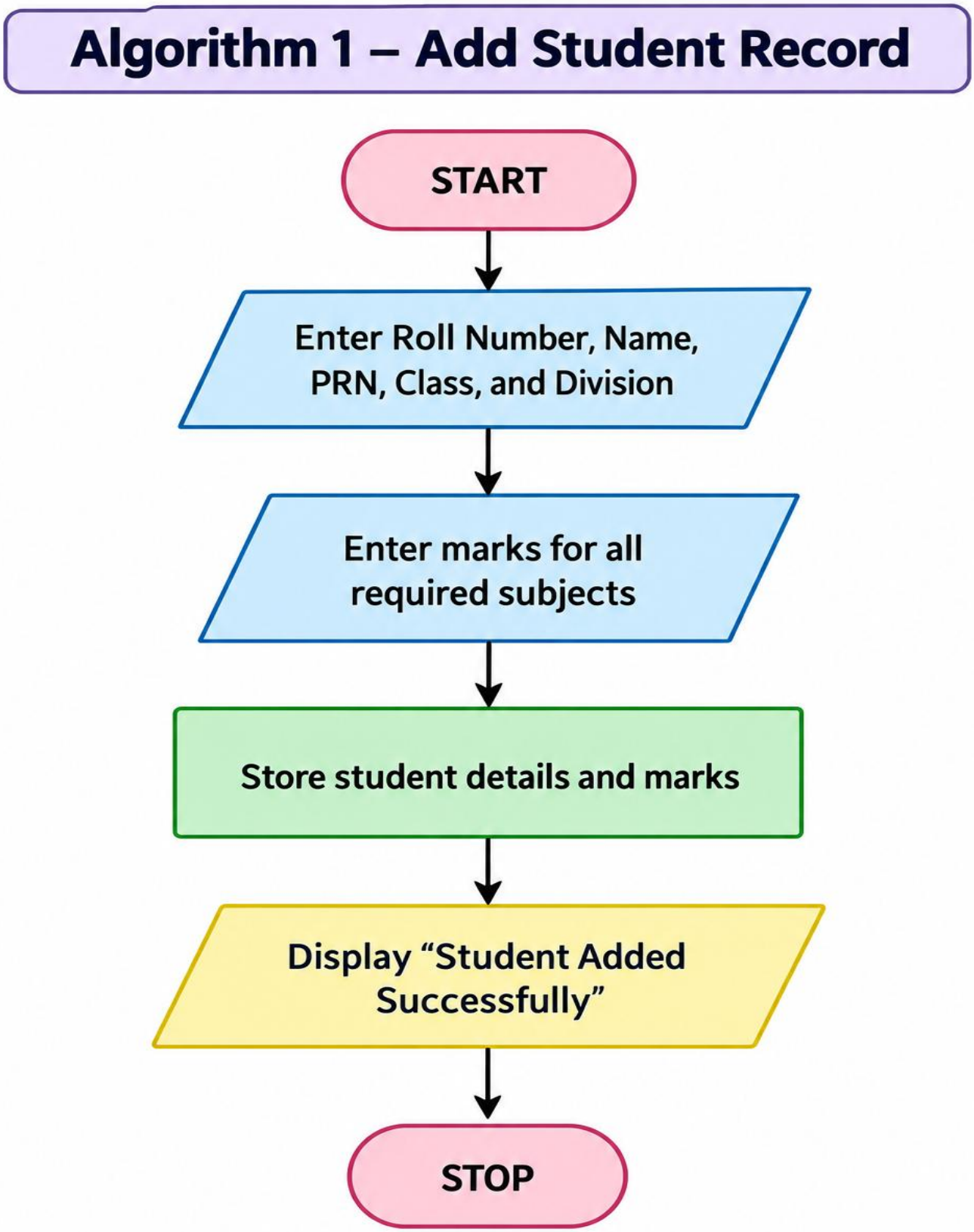
5.Compare the average with the grading rules.

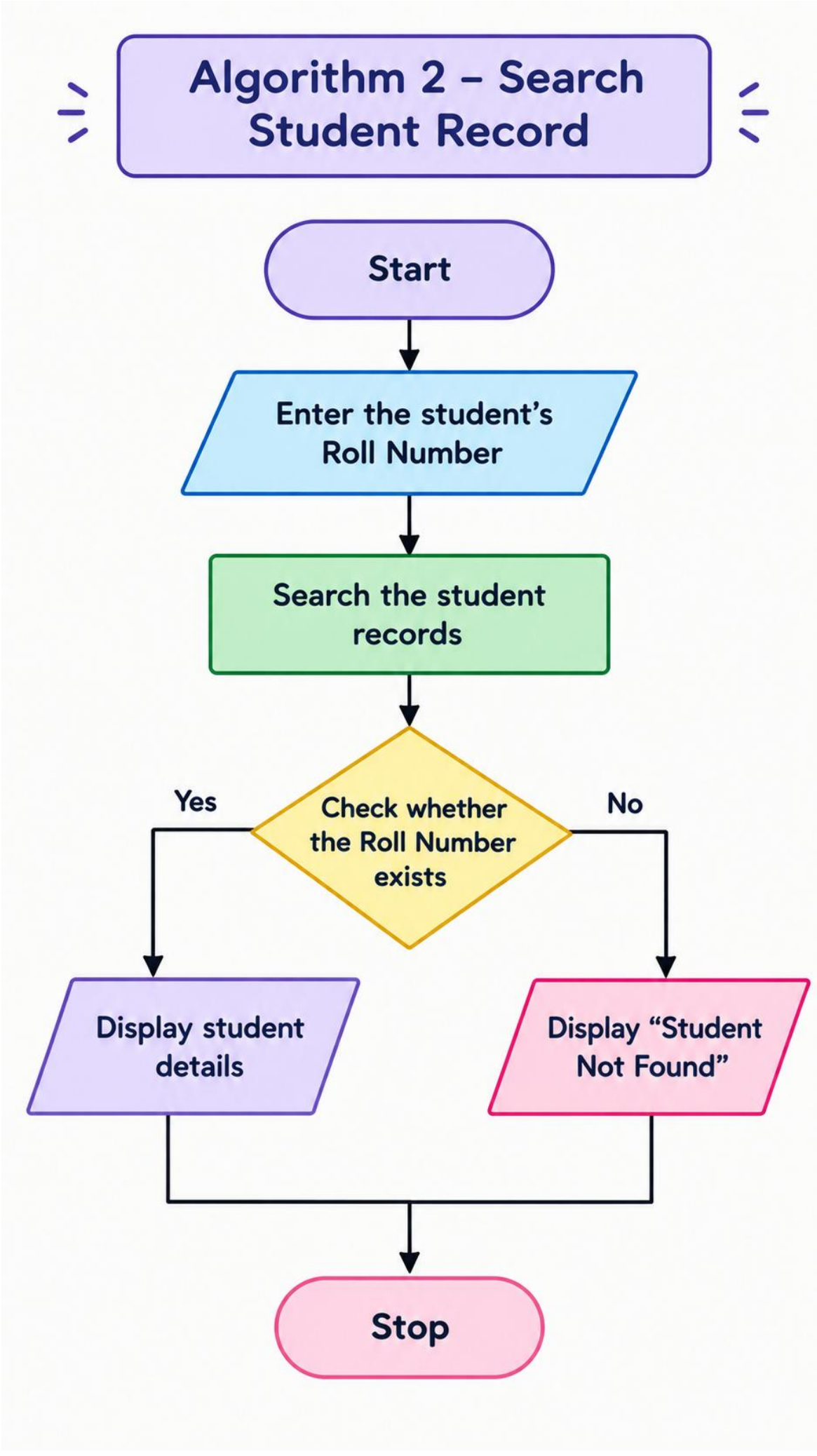
6.Assign the appropriate Grade.

7.Display Total Marks, Average Marks, and Grade.

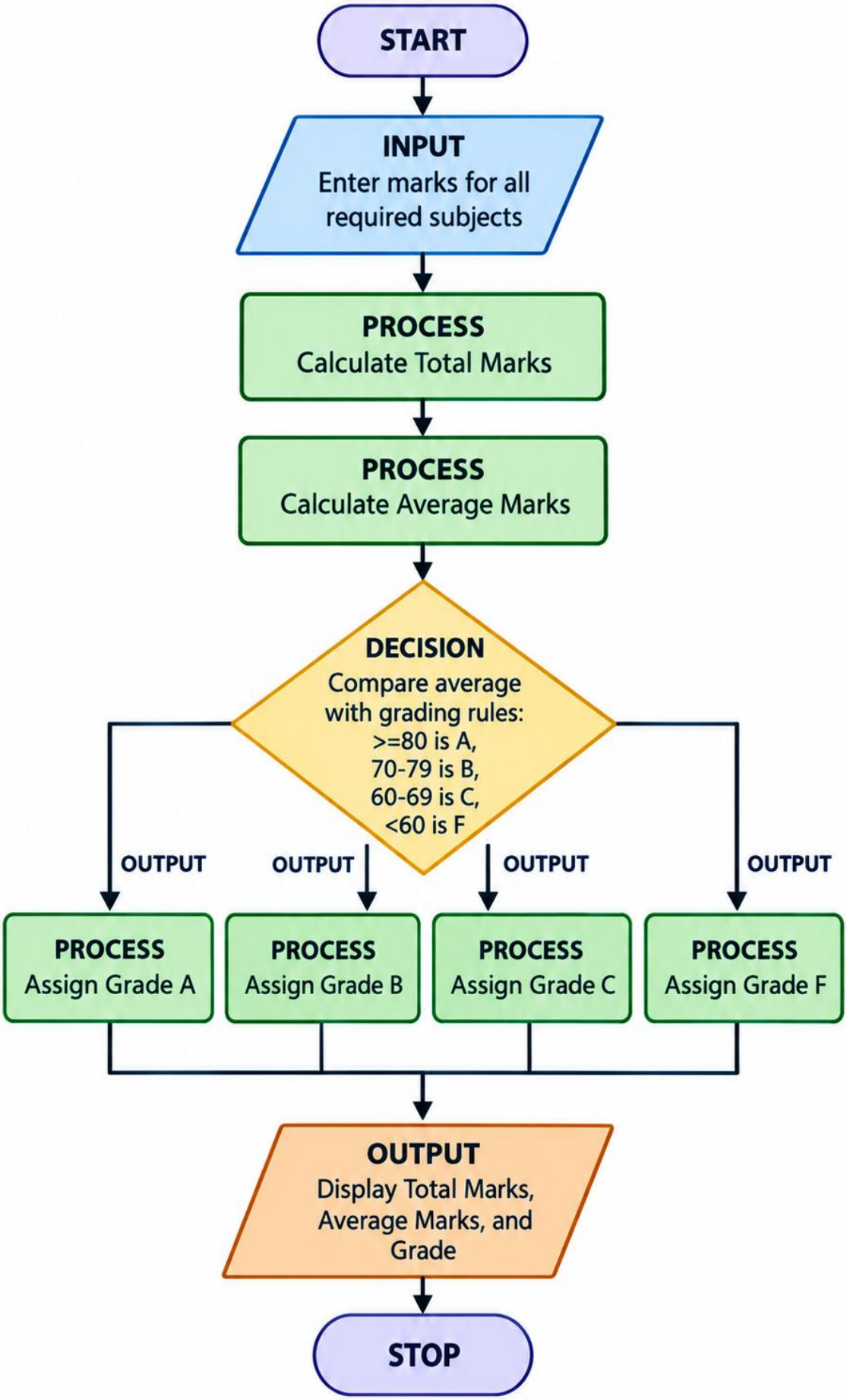
8.Stop.

Flowchart Collection




Algorithm 2 – Search Student Record
Start
Enter the student's Roll Number
Search the student records
Check whether the Roll Number exists
Yes
No
Display student details
Display "Student Not Found"
Stop

Algorithm 3 – Calculate Academic Result



Pseudocode Document

Pseudocode 1: Add Student Record

```
BEGIN
Input the student’s roll number.
Input the student’s name.
Input the student’s PRN.
Input the student’s class.
Input the student’s division.
Input the marks for all subjects.
Store the student’s information.
Display “Student Added Successfully”.
END
```

Pseudocode 2: Search Student Record

```
BEGIN
Input the student’s roll number.
Search the student records using the roll number.
If a matching record is found:
Display the student’s details.
Otherwise:
Display “Student Not Found”.
END
```

Pseudocode 3 – Calculate Average and Grade

```
BEGIN
    INPUT marks for all required subjects

    CALCULATE Total_Marks
    CALCULATE Average_Marks

    IF Average_Marks >= 80 THEN
        Grade = "A"
    ELSE IF Average_Marks >= 70 THEN
        Grade = "B"
    ELSE IF Average_Marks >= 60 THEN
        Grade = "C"
    ELSE
        Grade = "F"
    END IF

    DISPLAY Total_Marks
    DISPLAY Average_Marks
    DISPLAY Grade
END
```

Short Explanation of Proposed Solution

The proposed Student Record and Academic Management System is designed to provide a simple and organized way to manage student information and academic records. It will store important details such as Roll Number, Name, PRN, Class, Division, and subject-wise marks in one system.

The system will provide different operations such as Add Student, Search Student, View Student Details, Calculate Average Marks, and Calculate Grade. When adding a student, the user will enter the required personal and academic information, which will then be stored for future use.

The Search Student feature will allow the user to find a particular student's record using the Roll Number. If the record is available, the system will display the student's details; otherwise, it will show "Student Not Found."

The system will also process the marks entered for each subject. It will calculate the total marks and average marks and determine the student's grade according to the approved grading criteria. This reduces manual calculations and helps avoid common calculation errors.

Overall, the proposed system will make student record management faster, easier, organized, and more reliable. It can also provide a foundation for adding more features in the future, such as updating records, deleting records, generating reports, and storing data in a database.